Umang Goel

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Summary

Backend and database engineer with hands-on experience building scalable services and optimizing data workflows. Delivered a multi-mode SRT retention system that reclaimed 2GB of storage and cut query latency by 35%, and built enterprise REST APIs and Airflow ETL pipelines that processed 50K+ records per run with 25% lower latency. Strong in Java, Python, SQL, Docker, AWS, and CI/CD, with a focus on testing, automation, and reliable system performance to help build fast, scalable backend solutions.

EDUCATION

Arizona State University **Aug 2022 - May 2026**
Bachelor's, Computer Science *Tempe, AZ*

- **GPA:** 3.8/4.0
- **Coursework:** Data Structures & Algorithms, Software Engineering, Distributed Systems, Software Analysis & Design, Cloud Engineering, Object-Oriented Programming

TECHNICAL SKILLS

- **Languages & Frameworks**: Java, C#, Python, SQL, JavaScript, React, Flask, Spring (MVC exposure)
- **Backend, APIs & Databases**: RESTful APIs, Microservices, gRPC, PostgreSQL, Redis, RabbitMQ, ETL Pipelines, Caching Strategies
- **Cloud, DevOps & CI/CD**: AWS, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, CI/CD, Linux
- **Engineering Practices**: Object-Oriented Design, Design Patterns, SDLC, Agile/Scrum, TDD, Unit & Integration Testing, Code Reviews

PROFESSIONAL EXPERIENCE

DigiClips Media Analytics Platform **Aug 2025 - May 2026**
Backend & Database Engineer Intern (Capstone) *Tempe, AZ*

- Designed 6 MySQL stored procedures with REGEXP-based input validation (email, phone, ZIP) and centralized error logging, reducing invalid data ingestion errors by ~40% across 3 backend services
- Built a multi-mode SRT retention system with preview, execution, and reporting modes; enforced a 90-day cleanup policy, reclaiming ~2GB (~84%) of total database storage
- Optimized SRT ingestion and clip metadata queries using composite indexing and query rewrites, reducing p95 latency by ~35% under sustained load
- Deployed a Dockerized MySQL 8.x instance over OpenVPN; resolved production issues (duplicate procedure conflicts, DNS misconfigurations), maintaining 99%+ uptime over a 4-month sprint cycle

Honeywell and ASU Innovation Hub **Aug 2025 - Dec 2025**
Software Developer Extern *Tempe, AZ*

- Developed and maintained enterprise RESTful services in Java and Python, implementing scalable API design and optimizing request handling to reduce end-to-end latency by 25%
- Built Airflow ETL pipelines processing 50K+ records per pipeline execution using PostgreSQL and Redis, enabling reliable analytics ingestion
- Wrote 100+ unit and integration tests (xUnit, pytest), resolving 30+ defects with QA and reducing post-release issues by 10%

DMML Lab, Arizona State University **Dec 2024 - Jul 2025**
Software Developer Intern *Tempe, AZ*

- Developed distributed backend services using Python and Java, applying object-oriented design patterns and Agile practices to reduce p95 latency by 30%.
- Built event-driven pipelines with RabbitMQ handling 500+ events/min; designed data workflows processing 300K+ records with validation and concurrency controls, improving analytical accuracy by 15%
- Implemented TDD (75%+ coverage), CI/CD pipelines with Docker, and documentation standards, reducing production defects by 40% across 3 service modules

PUBLICATION

- Saketh Vishnubhatla, Alimohammad Beigi, Rui Heng Foo, Umang Goel, Ujun Jeong, Bohan Jiang, Adrienne Raglin, Huan Liu.An Interventional Approach to Real-Time Disaster Assessment via Causal Attribution..*ACM International Conference on Information and Knowledge Management (CIKM 2025), Demo Track.* https://dl.acm.org/doi/10.1145/3746252.3761487

PROJECTS

Real-Time Event Processing Engine **2025**

- Designed and implemented a high-throughput event processing system using Python services and Redis Streams, processing 10K+ events/sec across 4 worker nodes with consistent sub-50ms p95 latency.
- Implemented retry logic with 3-tier escalation, fault handling, and 60+ automated test cases to ensure 99.9% reliable event processing under peak load conditions.
- Monitored and triaged 15+ production-style incidents using structured logging and metrics dashboards, identifying root causes within 10 minutes and reducing recurring failures by 50%.

Causal MMD **2025**

- Designed scalable Flask application integrating 1.03M+ records with ensemble classifiers, achieving 92% accuracy for real-time disaster response coordination.
- Applied performance optimization through async I/O and batch processing with connection pooling, scaling throughput to 10K+ points/min while reducing latency by 30%.
- Built thread-safe REST API with versioned endpoints and OpenAPI docs, containerized with Docker and deployed via CI/CD with 95%+ test coverage.